Experiment No: 01
Experiment Name: Experiment on Basic Plotting using MATLAB Simulator

Objective:
i) To define and use variables and functions in MATLAB.
ii) To define and use vectors, matrics and polynomials in MATLAB.
iii) To study various MATLAB arithmetic operators and mathematics function.
iv) To create and use m-files.
v) Plot the basic signals.

Theory:
MATLAB is a programming language developed by Math Works. It started out as a matrix programming language where linear algebra programming was sample . MATLAB (matrix laboratory) is a fourth-generation high-level programming language and interactive environment for numerical computation, visualization and programming. MATLAB is developed by Math Works.
It allows matrix manipulations; plotting of functions and data; implementation of algorithms; creation of user interfaces; interfacing with programs written in other languages, including C, C++, Java and FORTRAN; analyze data; develop algorithms; and create models and applications.
It has numerous buit-in commands and math functions that help you in mathematical calculations, generating plots, and performing numerical methods.

Lab Practice 1: Enter the following matrix:

$$A = \begin{bmatrix} 10 & 60 & 90 \\ 30 & 40 & 55 \\ 50 & 20 & 10 \end{bmatrix}$$

Matlab Code:
```
clc
clear all;
close all;
A = [10 60 90; 30 40 55; 50 20 10]
```

Output:

```
A =

    10    60    90
    30    40    55
    50    20    10

>>
```

Lab Practice 2: Enter the following matrix:

$$A = \begin{bmatrix} 10 & 60 & 90 & 70 & 30 \\ 30 & 40 & 55 & 85 & 25 \\ 50 & 20 & 10 & 20 & 05 \\ 110 & 130 & 150 & 90 & 100 \end{bmatrix}$$

Matlab Code:

```
clc
clear all;
close all;
A = [10 60 90 70 30; 30 40 55 85 25; 50 20 10 20 05; 110 130 150 90 100]
X = A;
X(4,5) = 101;
```

Output:

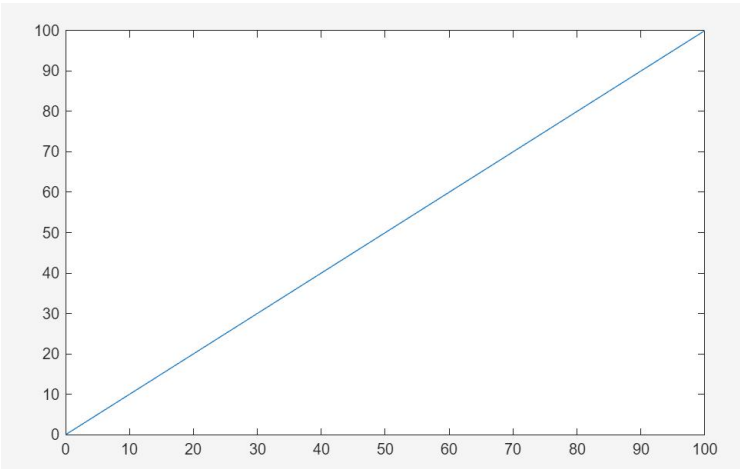
```
A =

    10    60    90    70    30
    30    40    55    85    25
    50    20    10    20     5
   110   130   150    90   100

>>
```

Lab Practice 3: Define x, by specifying the range of values for the variables x, for which the function is to be plotted.

Matlab Code:

```
clc
clear all;
close all;
x = [0:5:100];
y = x;
plot(x,y)
```

Output:



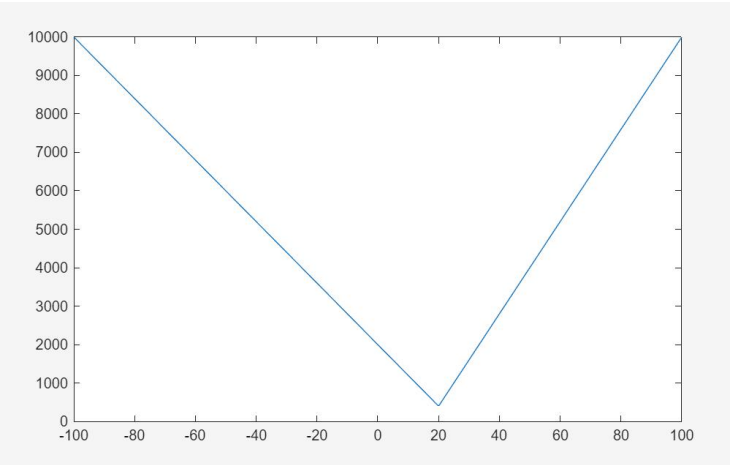
Lab Practice 4: To plot the function y=x^2.

Matlab Code:

```
clc
```

```
clear all;
close all;
x = [1 2 3 4 5 6 7 8 9 10];
x = [-100;20;100];
y = x.^2;
plot(x,y);
```

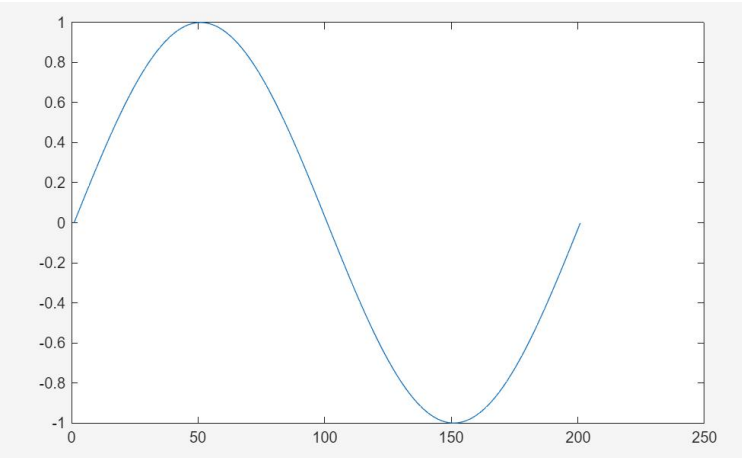
Output:



Lab Practice 5: To plot the y= sin(x).
Matlab Code:

```
clc
clear all;
close all;
x = 0:pi/100:2*pi;
y = sin(x);
plot(y);
```

Output:

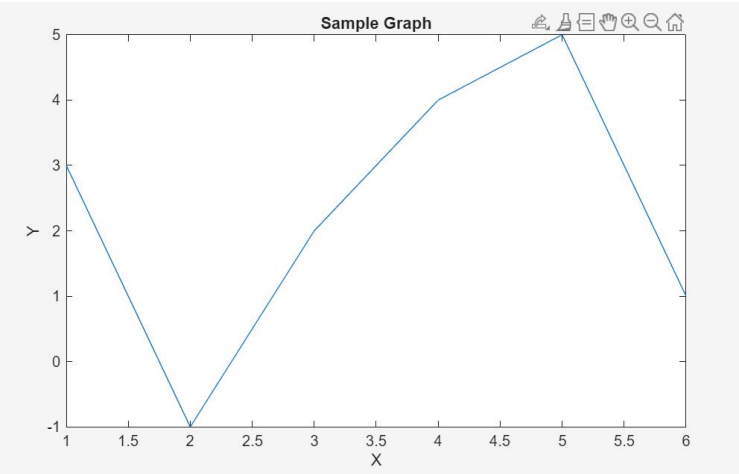


Lab Practice 6: Draw the simple graph according to given value: x=[1 2 3 4 5 6]; & y=[3 -1 2 4 5 1];
Matlab Code:

```
clc
clear all;
close all;
```

```
x=[1 2 3 4 5 6];
y=[3 -1 2 4 5 1];
plot(x,y);
xlabel ('X');
ylabel ('Y') ;
title('Sample Graph');
```

Output:

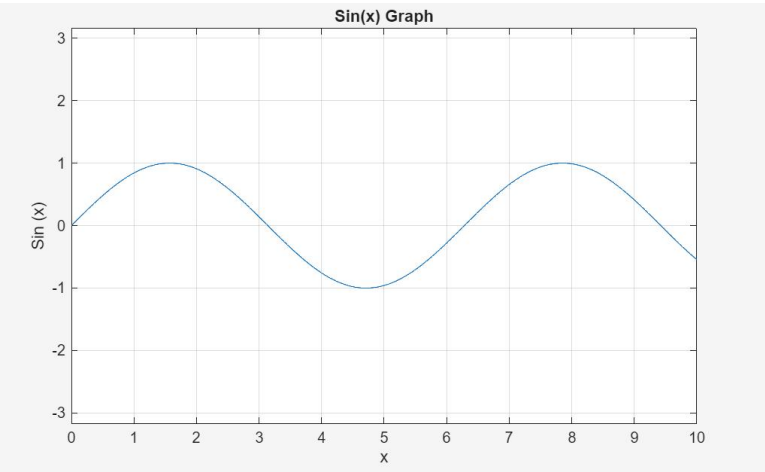


Lab Practice 7: Draw the y=sin(x) graph also to add title, tables along the x-axis and y=axis, grid lins and also to adjust the axes to spruce up the graph.

Matlab Code:

```
clc
clear all;
close all;
x=[0:0.01:10];
y=sin(x);
plot(x,y);
xlabel('x');
ylabel('Sin (x)');
title('Sin(x) Graph');
grid on;
axis equal
```

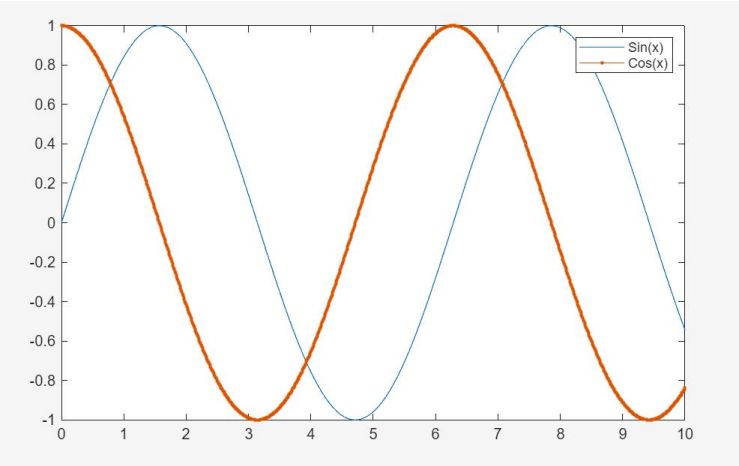
Output:



Lab Practice 8: Draw multiple graphs on the same plot.
Matlab Code :

```
clc
clear all;
close all;
x=[0:0.01:10];
y=sin(x);
g=cos(x);
plot(x, y, x, g, '.-');
legend('Sin(x)', 'Cos(x)')
```

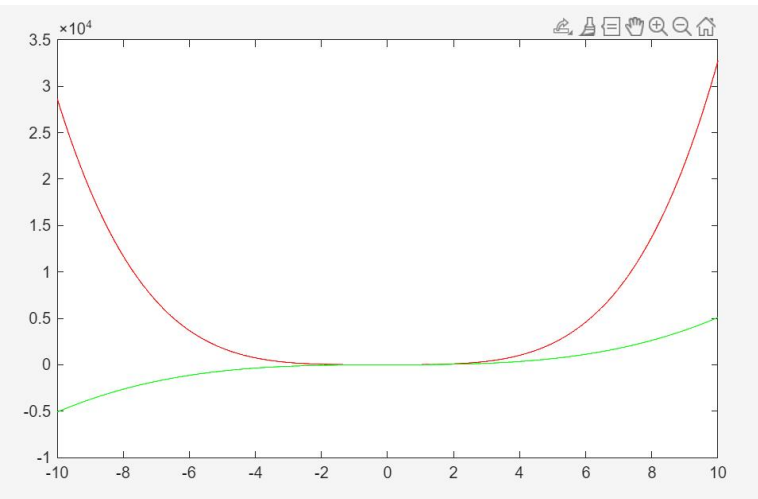
Output:



Lab Practice 9: Draw the graph of two polynomials $f(x)=3x^4 + 2x^3 +7x^2 + 2x + 9$ and $g(x) = 5x^3 + 9x +2$.
Matlab Code:

```
clc
clear all;
close all;
x=[-10:0.01:10];
y=3*x.^4 + 2 * x.^3 + 7 * x.^2 + 2 * x + 9;
g=5 * x.^3 + 9 * x + 2;
plot(x, y, 'r', x, g, 'g')
```

Output:

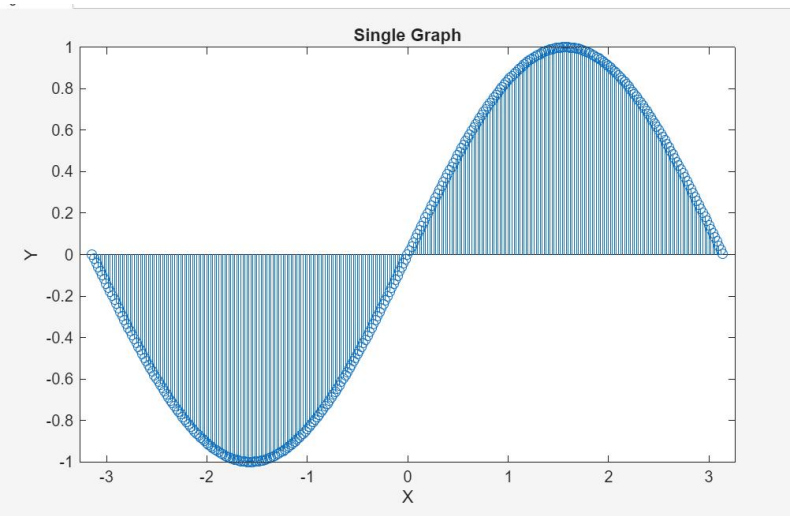


Lab Practice 10: Draw the stem graph of Y=sin(x)

Matlab Code:

```
clc
clear all;
close all;
X=[-pi:0.02:pi];
Y=sin(X);
stem(X,Y);
xlabel('X');
ylabel('Y');
title('Single Graph');
```

Output:

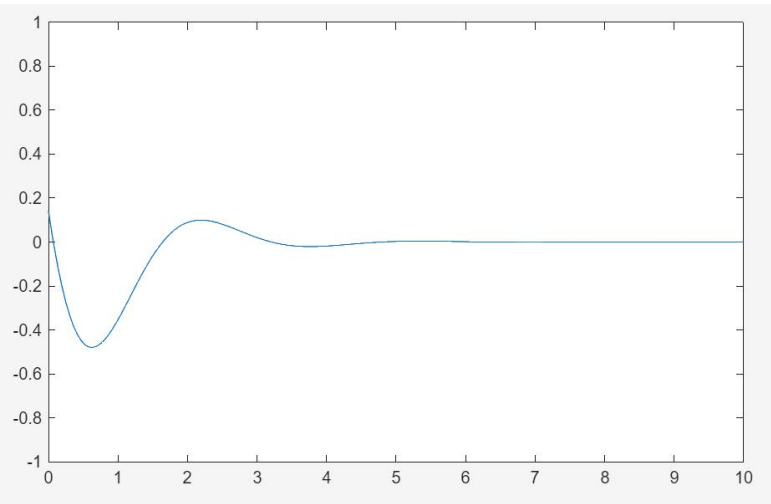


Lab Practice 11: Draw the Y= e^{-x} * sin(2x+3) graph.

Matlab Code:

```
clc
clear all;
close all;
x=[0:0.01:10];
y=exp(-x).* sin(2*x + 3);
plot(x,y),
axis([0 10 -1 1])
```

Output:

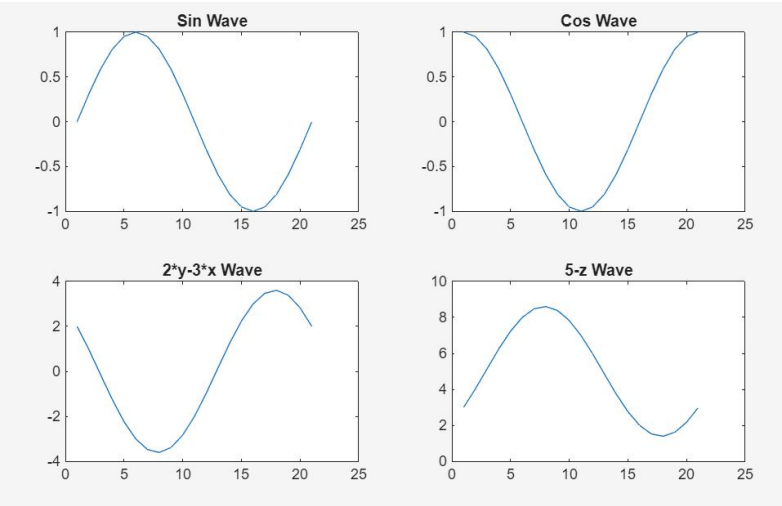


Lab Practice 12: Draw the x=sin(t); y=cos(t); z=2*y-3*x; v=5-z graph and shown in subplot.

Matlab Code:

```
clc
clear all;
close all;
t=0:pi/10:2*pi;
x=sin(t);
y=cos(t);
z=2*y-3*x;
v=5-z;
subplot(2,2,1);
plot(x)
title("Sin Wave")
subplot(2,2,2);
plot(y)
title("Cos Wave")
subplot(2,2,3);
plot(z)
title("2*y-3*x Wave")
subplot(2,2,4);
plot(v)
title("5-z Wave")
```

Output:

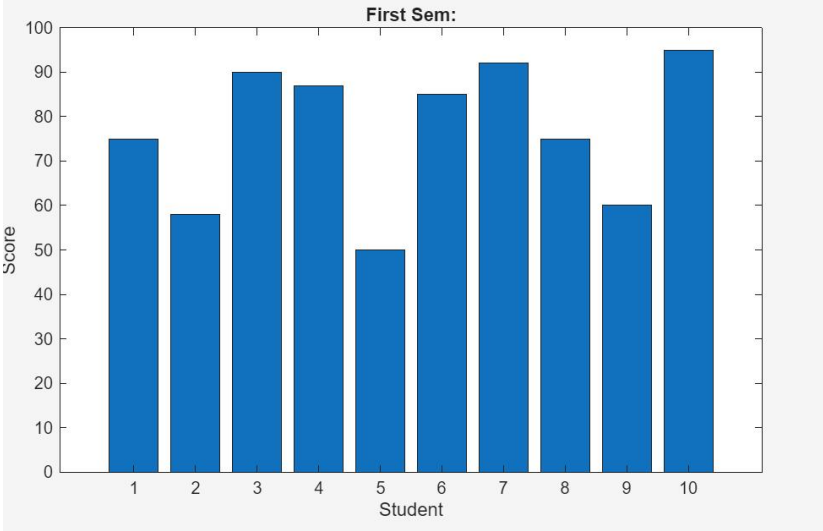


Lab Practice 13: Drawing Bar Charts.

Matlab Code:

```
clc
clear all;
close all;
x=[1:10];
y=[75, 58, 90, 87, 50, 85, 92, 75, 60, 95];
bar(x,y), xlabel('Student'),ylabel('Score'),
title('First Sem:')
```

Output:

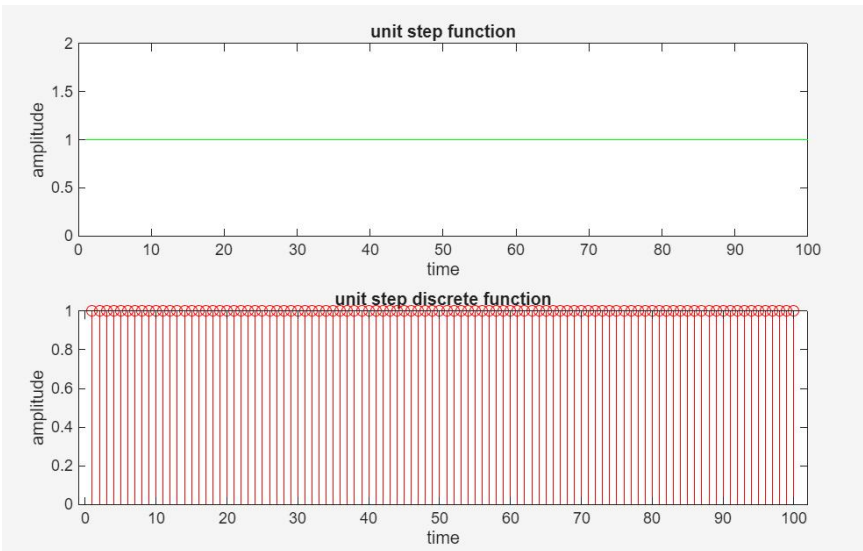


Lab Practice 14: Drawing Unit Step Function Graph for Continuous and Discrete Time.

Matlab Code:

```
clc
clear all;
close all;
N=100;
t=1:100;
x=ones(1,N);
subplot(2,1,1);
plot(t,x,'g');
xlabel('time');
ylabel('amplitude');
title('unit step function');
subplot(2,1,2);
stem(t,x,'r');
xlabel('time');
ylabel('amplitude');
title('unit step discrete function');
```

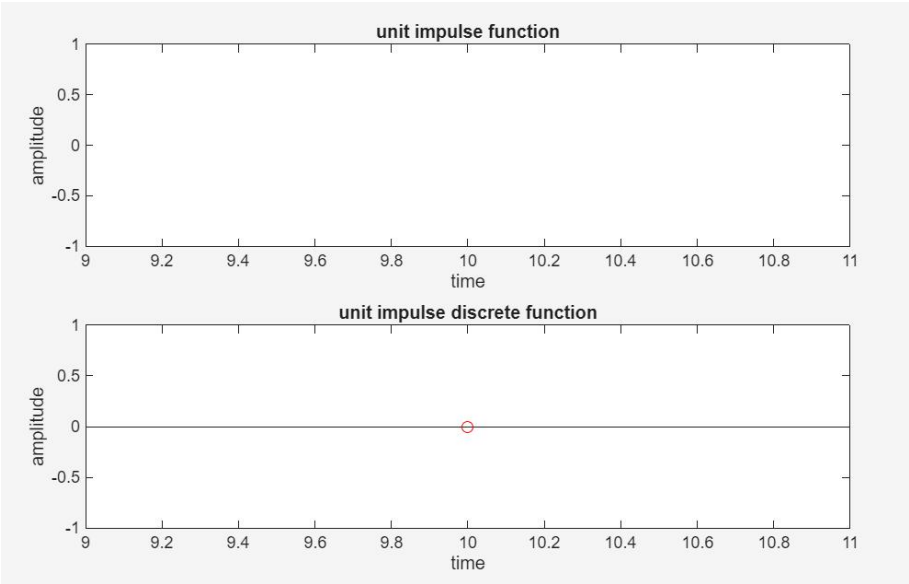
Output:



Lab Practice 15: Drawing Unit impulse Function Graph for Continuous and Discrete Time.
Matlab Code:

```
clc
clear all;
close all;
t=10:1:10;
x=(t==0);
subplot(2,1,1);
plot(t,x,'g');
xlabel('time');
ylabel('amplitude');
title('unit impulse function');
subplot(2,1,2);
stem(t,x,'r');
xlabel('time');
ylabel('amplitude');
title('unit impulse discrete function');
```

Output:

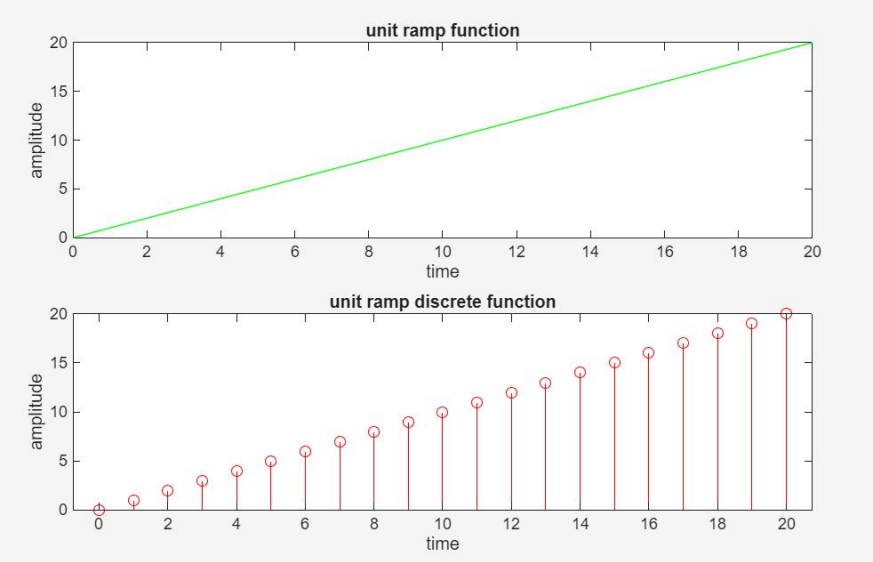


Lab Practice 16: Drawing Unit Ramp Function Graph for Continuous and Discrete Time.
Matlab Code:

```
clc
clear all;
close all;
t=0:0.01:2;
x=square(2*pi*t);
subplot(2,1,1);
plot(t,x,'g');
xlabel('time');
ylabel('amplitude');
title('unit ramp function');
subplot(2,1,2);
stem(t,x,'r');
xlabel('time');
ylabel('amplitude');
```

```
title('unit ramp discrete function');
```

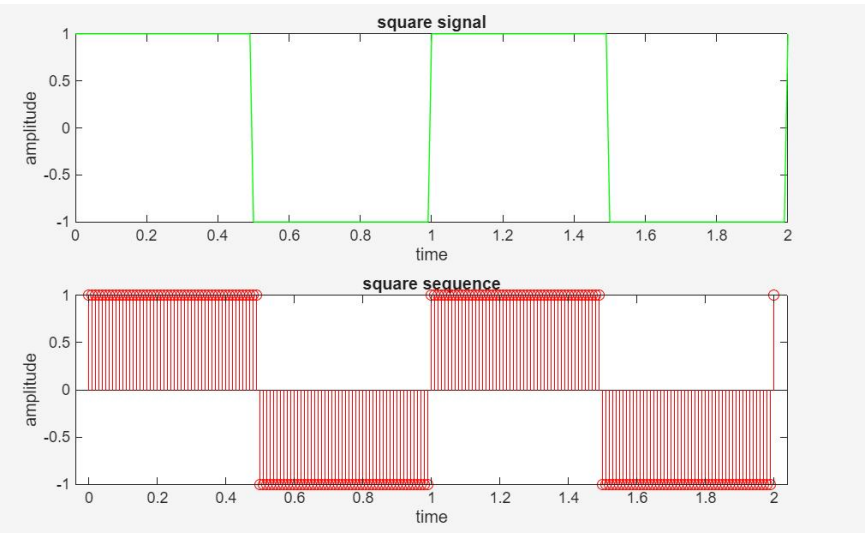
Output:



Lab Practice 17: Drawing Unit Square Function Graph for Continuous and Discrete Time.
Matlab Code:

```
clc
clear all;
close all;
t=0:0.01:2;
x=square(2*pi*t);
subplot(2,1,1);
plot(t,x,'g');
xlabel('time');
ylabel('amplitude');
title('square signal');
subplot(2,1,2);
stem(t,x,'r');
xlabel('time');
ylabel('amplitude');
title('square sequence');
```

Output:

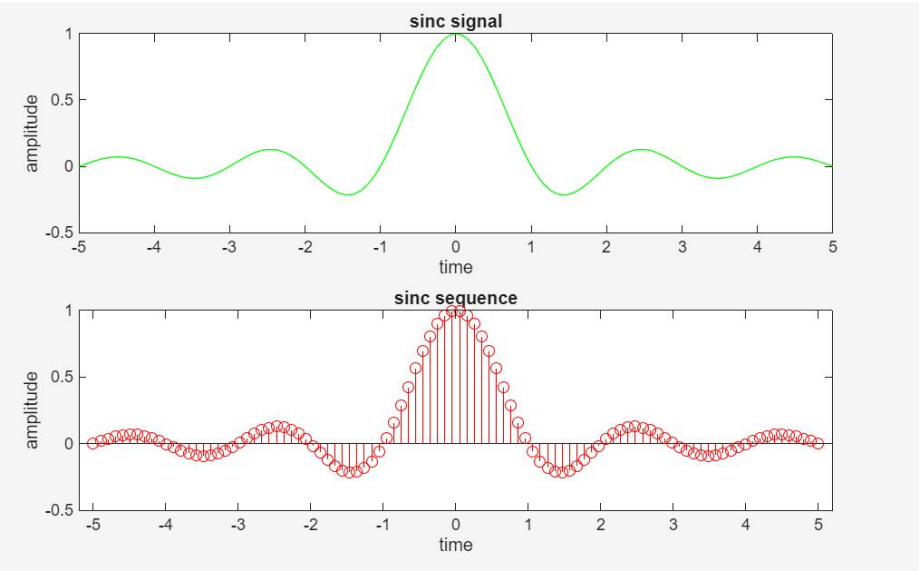


Lab Practice 18: Drawing Unit Sinc Function Graph for Continuous and Discrete Time.

Matlab Code:

```
clc
clear all;
close all;
t=linspace(-5,5);
x=sinc(t);
subplot(2,1,1);
plot(t,x,'g');
xlabel('time');
ylabel('amplitude');
title('sinc signal');
subplot(2,1,2);
stem(t,x,'r');
xlabel('time');
ylabel('amplitude');
title('sinc sequence');
```

Output:



Discussion: Through this experiment, we have learned how to create different types of graphs and plots using MATLAB. Specifically, we practiced using functions like plot(), xlabel(), ylabel(), title(), and legend() to represent data visually. We also learned how to plot multiple data sets on the same graph and how to enhance clarity through proper labeling and scaling.

This report will be useful in the future for analyzing and presenting data in various fields of engineering and science. MATLAB plotting is an essential skill for research, signal processing, control system design, and preparing technical reports. Visual representation of data makes complex information easier to understand, and this skill will be highly beneficial for our future academic projects as well as professional careers.